INCH-POUND

A–A–59590/5B
14 October 2005
SUPERSEDING
A-A-59590/5A
28 April 2005

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 3 PIECE, SCREW ACTUATED DRIVE, .225 X .260 INCH BODY SIZE, WITH SCREW RETENTION

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the latest issue in effect of A–A–59590.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).

AA59590	/05	B, C, H	*	* *	D, E, W
CID number	Applicable CID specification sheet	Finish	Mounting options	Length (see table I)	Options

Example: AA59590/05HT18 is the PIN for a hard black anodize finished, 1.8 inch (45.7 mm) long card holder with a retained actuating screw. The card holder also has two tapped mounting holes for use with 0–80 UNF fasteners.

SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card assembly firmly in its installed position. When the card holder is installed properly, it is capable of withstanding 60g/6ms of shock, 25 G-rms of vibration, and provides from 2 to 4 degrees C/W/inch thermal resistance transfer between the circuit card assembly and the heat sink surfaces.

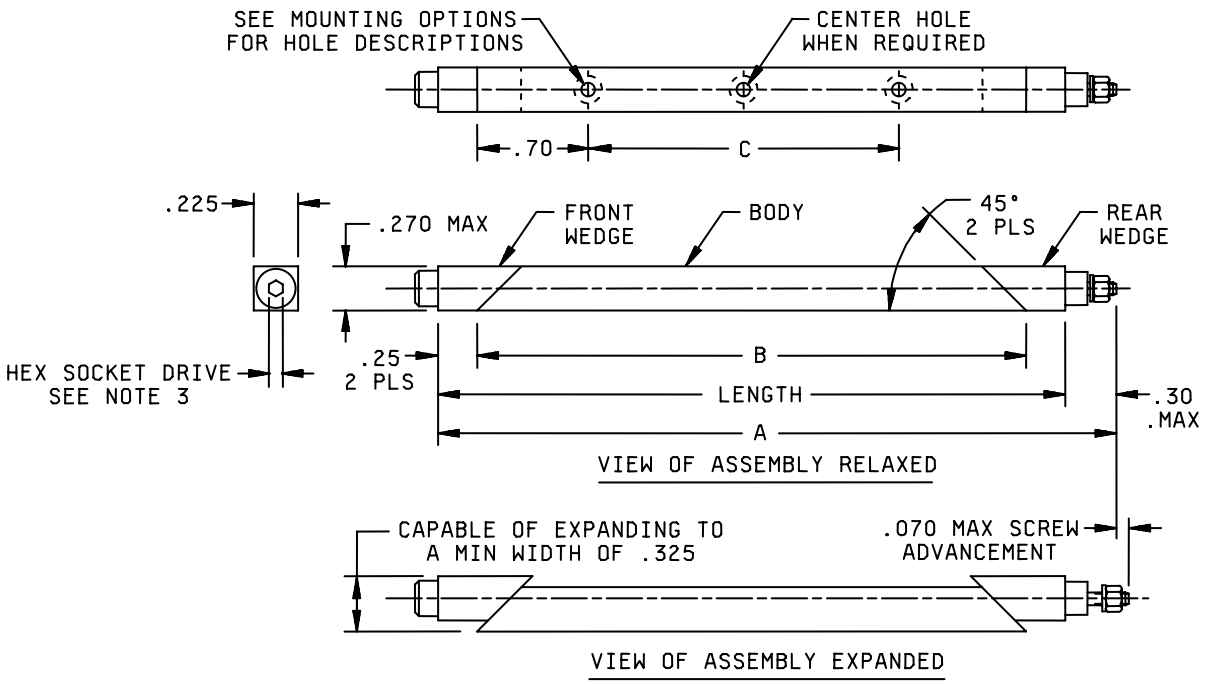
Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified on figures 1, 2, 3, 4, in table I, and A–A–59590.

Screw retention locknut (see figure 2). Card holders shall have a screw retention locknut intended to prevent unintentional disassembly of the card holder. The locknut shall be fabricated of a corrosion resistant material.

Nominal installation torque. When card holders are used in cold plate applications, the nominal installation torque of each card holder shall be as follows: 6 inch-pounds (0.68 N-m) for assemblies using no options or option “W” and 7.2 inch-pounds (0.81 N-m) for assemblies using options “E” or “D”.

Cold plate dimensions. Recommend cold plate slot width to accommodate the circuit card assembly with attached card holder is .300 inch (7.62 mm) plus the thickness of the printed board of the circuit card assembly (see A–A–59590).

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.070	1.78	.250	6.35	.270	6.86	.325	8.26
.225	5.72	.25	6.35	.30	7.6	.70	17.8

NOTES:

1. Dimensions are in inches. Millimeters are given for information only.
2. Unless otherwise specified, tolerances are for ± .02 inch (0.51 mm) for two place decimals and ± .010 (0.25 mm) for three place decimals.
3. The across flats dimension for hex drive socket shall be .094 inch (2.38 mm) for mounting options “J”, “K”, “N”, “T”, “R”, “V”, and “S”. The across flats dimension for hex drive socket shall be a metric dimension of 2.5 mm (0.98 inch) across flats for mounting options “G” and “M”.

FIGURE 1. Relaxed and expanded dimensions.



FIGURE 2. Screw retaining self-locking nut (corrosion resistant) details.

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TABLE I. Additional assembly dimensions (see figure 1). 1/

PIN length designator	Dimension "LENGTH" ± .02 (0.5)	Dimension "A" +.02 / –.09 (+0.5 / –2.3)		Dimension "B" ± .03 (0.8)	Dimension "C" ± .005 (0.13)
		Without options W or D	With options W or D		
18	1.8 (45.7)	2.10 (53.3)	2.20 (55.9)	1.30 (33.0)	.75 (19.1)
23	2.3 (58.4)	2.60 (66.0)	2.70 (68.6)	1.80 (45.7)	.40 (10.2)
28	2.8 (71.1)	3.10 (78.7)	3.20 (81.3)	2.30 (58.4)	.90 (22.9)
30	3.0 (76.2)	3.30 (83.8)	3.40 (86.4)	2.50 (63.5)	1.10 (27.9)
33	3.3 (83.8)	3.60 (91.4)	3.70 (94.0)	2.80 (71.1)	1.40 (35.6)
35	3.5 (88.9)	3.80 (96.5)	3.90 (99.1)	3.00 (76.2)	1.60 (40.6)
38	3.8 (96.5)	4.10 (104.1)	4.20 (106.7)	3.30 (83.8)	1.90 (48.3)
43	4.3 (109.2)	4.60 (116.8)	4.70 (119.4)	3.80 (96.5)	2.40 (61.0)
48	4.8 (121.9)	5.10 (129.5)	5.20 (132.1)	4.30 (109.2)	2.90 (73.7)
53	5.3 (134.6)	5.60 (142.2)	5.70 (144.8)	4.80 (121.9)	3.40 (86.4)
55	5.5 (139.7)	5.80 (147.3)	5.90 (149.9)	5.00 (127.0)	3.60 (91.4)
58	5.8 (147.3)	6.10 (154.9)	6.20 (157.5)	5.30 (134.6)	3.90 (99.1)
61	6.1 (154.9)	6.40 (162.6)	6.50 (165.1)	5.60 (142.2)	4.20 (106.7)
63	6.3 (160.0)	6.60 (167.6)	6.70 (170.2)	5.80 (147.3)	4.40 (111.8)
67	6.7 (170.2)	7.00 (177.8)	7.10 (180.3)	6.20 (157.5)	4.80 (121.9)
68	6.8 (172.7)	7.10 (180.3)	7.20 (182.9)	6.30 (160.0)	4.90 (124.5)
73	7.3 (185.4)	7.60 (193.0)	7.70 (195.6)	6.80 (172.7)	5.40 (137.2)
78	7.8 (198.1)	8.10 (205.7)	8.20 (208.3)	7.30 (185.4)	5.90 (149.9)
83	8.3 (210.8)	8.60 (218.4)	8.70 (221.0)	7.80 (198.1)	6.40 (162.6)
88	8.8 (223.5)	9.10 (231.1)	9.20 (233.7)	8.30 (210.8)	6.90 (175.3)

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for information only.

Finish. The finish options shall be as specified in A–A–59590. The finish options available for this specification sheet are as follows: "B" (black anodize), "C" (chemical film), or "H" (hard black anodize).

Mounting options. The mounting options shall be as specified in A–A–59590. The options available for this specification sheet are as follows: "J" or "K" (rivet mounting holes), "N" (no mounting holes), "T" or "R" (tapped 0-80 UNF 2B holes), "V" or "S" (tapped 2-56 UNC 2B holes), or "G" or "M" (tapped M2.5 x.45 metric holes). Card holders using mounting option "J" or "K" may be shipped unassembled.

Mounting holes (when required). Two mounting holes are required on card holders less than 5.30 inches (134.6 mm) in length. Three mounting holes are required on card holders 5.30 inches (134.6 mm) or greater in length. The third mounting hole shall be centered on the mounting body. See figure 1 and table I for mounting hole locations and spacing requirements.

Rivet mounting holes. The holes used for rivet mounting shall be .066 to .073 inch (1.68 to 1.85 mm) diameter through holes, countersunk 100 degrees by .140 inch (3.56 mm) diameter with an access/clearance counterbore hole of .156 inch (3.96 mm) diameter by .200 inch (5.08 mm) deep.

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Length, expanded, and relaxed dimensions. The length, expanded, and relaxed dimensions shall be as specified on figure 1. The length designator shall be as specified in A–A–59590 and the available lengths for this specification sheet are listed in table I.

Options. Card holders can have the following options. Card holders not requiring the following option shall leave the options position in the PIN blank.

Lockwasher and flat washer (see figure 3). A lockwasher and flat washer located under the screw head will provide for additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flat washer option shall include a suffix “W” in the PIN (see classification and notes).

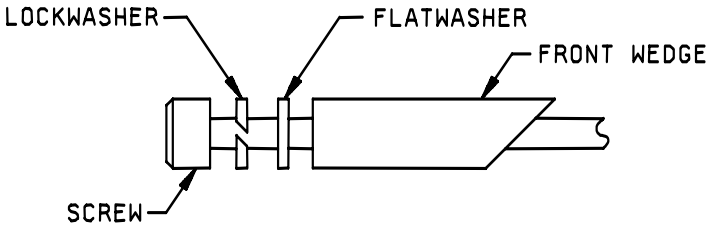


FIGURE 3. Lockwasher and flat washer details.

Screw self-locking element (see figure 4). The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock and vibration. Card holders requiring a screw self-locking element shall include a suffix “E” in the PIN (see classification and notes).



FIGURE 4. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element option shall include a suffix “D” in the PIN (see classification and notes).

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A–A–59590 – Holder, Electrical Card, Wedge Retainers, 3 Piece, Screw Actuated Drive, General Requirements For.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094.)

Ordering data. Ordering data is as specified in A–A–59590.

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Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
18915	APW Electronic Solutions 14100 Danielson Street Poway, CA 92064–6898	Telephone: (858) 679–4550 Facsimile: (858) 679–4555 Electronic mail: sales.electronicsolutions@apw.com Uniform Resource Locator (URL): www.apw.com
52094	Calmark Corporation 4915 Walnut Grove Avenue San Gabriel, CA 91776–2099	Telephone: (626) 287–0451 Facsimile: (626) 287–7350 Electronic mail: sales@calmark.com URL: www.calmark.com

Part number supersession data. These CID specification sheet part numbers supersede the following manufacturer's part numbers as shown in table II. The CID PINs listed in table II are only for length designator "28". See table III for CID PIN construction using other available lengths for this specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. Commercial part number supersession data.

PIN designator AA59590/05	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/05	Vendor similar designator or type part number 1/ CAGE 52094
CJ28	40–6–LN	CJ28	225CR2.80HR
CN28	40–6–LN–A	CN28	225CR2.80R
CT28	40–6–LN–T	CT28	225CR2.80T0R
CV28	40–6–LN–S	CV28	225CR2.80T2R
CG28	40–6–LN–M	CG28	M225CR2.80TM2.5R
CJ28E	40–6–LN–L	CJ28E	225CR2.80HLR
CN28E	40–6–LN–A–L	CN28E	225CR2.80LR
CT28E	40–6–LN–T–L	CT28E	225CR2.80T0LR
CV28E	40–6–LN–S–L	CV28E	225CR2.80T2LR
CG28E	40–6–LN–M–L	CG28E	M225CR2.80TM2.5LR
CJ28W	40–6–LF–LN	CJ28W	W225CR2.80HR
CN28W	40–6–LF–LN–A	CN28W	W225CR2.80R
CT28W	40–6–LF–LN–T	CT28W	W225CR2.80T0R
CV28W	40–6–LF–LN–S	CV28W	W225CR2.80T2R
CG28W	40–6–LF–LN–M	CG28W	MW225CR2.80TM2.5R

See footnote at end of table.

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TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/05	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/05	Vendor similar designator or type part number 1/ CAGE 52094
CJ28D	40–6–LF–LN–L	CJ28D	W225CR2.80HLR
CN28D	40–6–LF–LN–A–L	CN28D	W225CR2.80LR
CT28D	40–6–LF–LN–T–L	CT28D	W225CR2.80T0LR
CV28D	40–6–LF–LN–S–L	CV28D	W225CR2.80T2LR
CG28D	40–6–LF–LN–M–L	CG28D	MW225CR2.80TM2.5LR
BJ28	40–6–B–LN	BJ28	A225CR2.80HR
BN28	40–6–B–LN–A	BN28	A225CR2.80R
BT28	40–6–B–LN–T	BT28	A225CR2.80T0R
BV28	40–6–B–LN–S	BV28	A225CR2.80T2R
BG28	40–6–B–LN–M	BG28	MA225CR2.80TM2.5R
BJ28E	40–6–B–LN–L	BJ28E	A225CR2.80HLR
BN28E	40–6–B–LN–A–L	BN28E	A225CR2.80LR
BT28E	40–6–B–LN–T–L	BT28E	A225CR2.80T0LR
BV28E	40–6–B–LN–S–L	BV28E	A225CR2.80T2LR
BG28E	40–6–B–LN–M–L	BG28E	MA225CR2.80TM2.5LR
BJ28W	40–6–B–LF–LN	BJ28W	WA225CR2.80HR
BN28W	40–6–B–LF–LN–A	BN28W	WA225CR2.80R
BT28W	40–6–B–LF–LN–T	BT28W	WA225CR2.80T0R
BV28W	40–6–B–LF–LN–S	BV28W	WA225CR2.80T2R
BG28W	40–6–B–LF–LN–M	BG28W	MWA225CR2.80TM2.5R
BJ28D	40–6–B–LF–LN–L	BJ28D	WA225CR2.80HLR
BN28D	40–6–B–LF–LN–AL	BN28D	WA225CR2.80LR
BT28D	40–6–B–LF–LN–TL	BT28D	WA225CR2.80T0LR
BV28D	40–6–B–LF–LN–SL	BV28D	WA225CR2.80T2LR
BG28D	40–6–B–LF–LN–ML	BG28D	MWA225CR2.80TM2.5LR
HJ28	40–6–B3–LN	HJ28	HA225CR2.80HR
HN28	40–6–B3–LN–A	HN28	HA225CR2.80R
HT28	40–6–B3–LN–T	HT28	HA225CR2.80T0R
HV28	40–6–B3–LN–S	HV28	HA225CR2.80T2R
HG28	40–6–B3–LN–M	HG28	MHA225CR2.80TM2.5R

See footnote at end of table.

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TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/05	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/05	Vendor similar designator or type part number 1/ CAGE 52094
HJ28E	40–6–B3–LN–L	HJ28E	HA225CR2.80HLR
HN28E	40–6–B3–LN–A–L	HN28E	HA225CR2.80LR
HT28E	40–6–B3–LN–T–L	HT28E	HA225CR2.80T0LR
HV28E	40–6–B3–LN–S–L	HV28E	HA225CR2.80T2LR
HG28E	40–6–B3–LN–M–L	HG28E	MHA225CR2.80TM2.5LR
HJ28W	40–6–B3–LF–LN	HJ28W	WHA225CR2.80HR
HN28W	40–6–B3–LF–LN–A	HN28W	WHA225CR2.80R
HT28W	40–6–B3–LF–LN–T	HT28W	WHA225CR2.80T0R
HV28W	40–6–B3–LF–LN–S	HV28W	WHA225CR2.80T2R
HG28W	40–6–B3–LF–LN–M	HG28W	MWHA225CR2.80TM2.5R
HJ28D	40–6–B3–LF–LN–L	HJ28D	WHA225CR2.80HLR
HN28D	40–6–B3–LF–LNAL	HN28D	WHA225CR2.80LR
HT28D	40–6–B3–LF–LNTL	HT28D	WHA225CR2.80T0LR
HV28D	40–6–B3–LF–LNSL	HV28D	WHA225CR2.80T2LR
HG28D	40–6–B3–LF–LNML	HG28D	MWHA225CR2.80TM2.5LR

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.

TABLE III. Example of PIN with available length designators.

PIN designator AA59590/05	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/05	Vendor similar designator or type part number 1/ 2/ CAGE 52094
HT18	40–4–B3–LN	HT18	HA225CR1.80T0R
HT23	40–5–B3–LN	HT23	HA225CR2.30T0R
HT28	40–6–B3–LN	HT28	HA225CR2.80T0R
HT30	3/	HT30	HA225CR3.00T0R
HT33	40–7–B3–LN	HT33	HA225CR3.30T0R
HT35	3/	HT35	HA225CR3.50T0R
HT38	40–8–B3–LN	HT38	HA225CR3.80T0R

See footnotes at end of table.

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TABLE III. Example of PIN with available length designators – Continued.

PIN designator AA59590/05	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/05	Vendor similar designator or type part number 1/ 2/ CAGE 52094
HT43	40–9–B3–LN	HT43	HA225CR4.30T0R
HT48	40–10–B3–LN	HT48	HA225CR4.80T0R
HR53	40–11–B3–LN	HR53	HA225CR5.30ET0R
HR55	3/	HR55	HA225CR5.50ET0R
HR58	40–12–B3–LN	HR58	HA225CR5.80ET0R
HR61	3/	HR61	HA225CR6.10ET0R
HR63	40–13–B3–LN	HR63	HA225CR6.30ET0R
HR67	3/	HR67	HA225CR6.70ET0R
HR68	40–14–B3–LN	HR68	HA225CR6.80ET0R
HR73	40–15–B3–LN	HR73	HA225CR7.30ET0R
HR78	40–16–B3–LN	HR78	HA225CR7.80ET0R
HR83	40–17–B3–LN	HR83	HA225CR8.30ET0R
HR88	40–18–B3–LN	HR88	HA225CR8.80ET0R

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.
2/ Other lengths are available on request.
3/ Consult manufacturer for part number.

MILITARY INTERESTS:

Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA – CC

Review activity:
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FSS

Preparing Activity
DLA – CC

Project 5998–2005–014

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